Real and Complex Numbers of Binomial Coefficients in Combinatorial Geometric Series

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Abstract: This paper discusses the real and complex numbers of binomial coefficients in combinatorial geometric series. The coefficient for each term in combinatorial geometric series refers to a binomial coefficient. This idea can enable the scientific researchers to solve the real life problems.

MSC Classification codes: 05A10, 40A05 (65B10)

Keywords: computation**,** combinatorics, binomial coefficient

1. Introduction

When the author of this article was trying to develop the multiple summations of geometric series, a new idea was stimulated his mind to create a combinatorial geometric series [1-9]. The combinatorial geometric series is a geometric series whose coefficient of each term of the geometric series denotes the binomial coefficient V_n^r. In this article, we show that the binomial coefficients in combinatorial geometric are a system of natural numbers.

2. Combinatorial Geometric Series

The combinatorial geometric series [1-9] is derived from the multiple summations of geometric series. The coefficient of each term in the combinatorial refers to the binomial coefficient V_n^r.

$$\sum_{i_1=0}^{n}\sum_{i_2=i_1}^{n}\sum_{i_3=i_2}^{n}\cdots\sum_{i_r=i_{r-1}}^{n} x^{i_r}=\sum_{i=0}^{n} V_i^r x^i \;\&\; V_n^r=\frac{(n+1)(n+2)(n+3)\cdots(n+r-1)(n+r)}{r!},$$

where $n\geq 0, r\geq 1$ & $n,r\in N$. Here, $N=\{1,2,3,4,5,\cdots\}$ is a system of natural numbers.

$\sum_{i=0}^{n} V_i^r x^i$ refers to the combinatorial geometric series;

V_n^r is the binomial coefficient of combinatorial geometric series and

$V_n^r=\frac{(n+r)!}{n!\,r!}$ is a positive integer (say, k), $i.e.\,(n+r)!=k\times n!\times r!$.

Here, $V_0^1=1;\ V_1^1=2;\ V_2^1=3;\ V_3^1=4;\ V_4^1=5;\ V_5^1=6;\ \cdots$

$N=\{V_0^1,V_1^1,V_2^1,V_3^1,\ V_4^1,V_4^1,\cdots\}$ is a set of natirual umbers .

$W=\{0,V_0^1,V_1^1,V_2^1,V_3^1,\ V_4^1,V_4^1,\cdots\}$ is a system of whole umbers .

$Z=\{\cdots,-V_2^1,-V_1^1,-V_0^1,0,V_0^1,V_1^1,V_2^1,\cdots\}$ is a system of integers.

If $x = 1$ in the combinatorial geometric series, then $\sum_{i=0}^{n} V_i^r x^i = \sum_{i=0}^{n} V_i^r$,

$i.e.\ V_0^r + V_1^r + V_2^r + V_3^r + \cdots + V_{n-1}^r + V_n^r = V_n^{r+1} \implies V_{n-1}^{r+1} + V_n^r = V_n^{r+1}$.

If r is either a real or complex number, then $\sum_{i=0}^{n} rV_i^r = rV_n^{r+1}$.

Then, $\sum_{i=0}^{n} rV_i^r x^i = \sum_{i=0}^{n} \mu_i x^i$, $(rV_i^r = \mu_i \text{ for } i = 0, 1, 2, 3, \cdots)$,

where μ_i, (i = 0, 1, 2, 3, . . .), are real or complex numbers and x is a real or complex variable.

3. Conclusion

In this article, the binomial coefficients of combinatorial geometric series were showed as real and complex numbers. This new idea can enable the scientific researchers for research and development further.

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